5. Indices and Logarithms

Laws of indices:

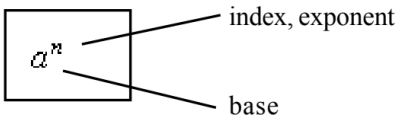
a, b **are real numbers.** m **and** n **are rational numbers.**

$$a^m \times a^n = a^{m+n}$$
$$a^m \div a^n = a^{m-n}$$
$$(ab)^m = a^m b^m$$
$$(a^m)^n = a^{mn}$$

When $a \neq 0$, **and** **is a rational number.**

$$a^{-n} = \frac{1}{a^n} \quad \text{and} \quad a^0 = 1$$

In a^n a **is called the base and** n **the index or exponent.**



An equation in which the variable is an exponent is called exponential equation.
For example $2^x = 32$ **is an exponential equation.**

Example 1
Find the values of the following when $x = 9$ **and** $y = 16$.

(a) $x^{\frac{1}{2}}.y^{\frac{3}{4}}$ (b) $\left(\frac{6x}{y}\right)^{\frac{1}{3}}$ (c) $(4xy)^{-\frac{1}{2}}$ (d) $(x+y)^{-\frac{1}{2}}$

(a) $x^{\frac{1}{2}}.y^{\frac{3}{4}} = 9^{\frac{1}{2}} \times 16^{\frac{3}{4}}$
$= \left(3^2\right)^{\frac{1}{2}} \times \left(2^4\right)^{\frac{3}{4}}$
$= 3 \times 2^3 = 3 \times 8 = 24$

(b) $\left(\frac{6x}{y}\right)^{\frac{1}{3}} = \left(\frac{6 \times 9}{16}\right)^{\frac{1}{3}} = \left(\frac{27}{8}\right)^{\frac{1}{3}} = \left[\left(\frac{3}{2}\right)^3\right]^{\frac{1}{3}} = \frac{3}{2}$

(c) $(4xy)^{-\frac{1}{2}} = (4 \times 9 \times 16)^{-\frac{1}{2}} = \left[(2 \times 3 \times 4)^2\right]^{-\frac{1}{2}} = (2 \times 3 \times 4)^{-1} = 24^{-1} = \frac{1}{24}$

(d) $(x+y)^{-\frac{1}{2}} = (9+16)^{-\frac{1}{2}} = 25^{-\frac{1}{2}} = (5^2)^{-\frac{1}{2}} = 5^{-1} = \frac{1}{5}$

Example 2

Solve: (a) $2^x = 10^3 \times 5^{-x}$

(b) $16^{x-1} = \frac{1}{8}$

(a)
$$2^x = 10^3 \times 5^{-x}$$
$$2^x = 10^3 \times \frac{1}{5^x}$$
$$2^x \times 5^x = 10^3$$
$$10^x = 10^3$$
$$x = 3$$

(b)
$$16^{x-1} = \frac{1}{8}$$
$$(2^4)^{x-1} = \frac{1}{2^3}$$
$$2^{4x-4} = 2^{-3}$$
$$4x - 4 = -3$$
$$4x = 1$$
$$x = \frac{1}{4}$$

Logarithm

Consider $y = 3^x$. **It will be observed that** y **must be positive for all real values of** x.

When $x = 2$, $y = 9$
$x = 3$, $y = 27$
$x = 0$, $y = 1$
$x = -4$, $y = \frac{1}{81}$

In $3^x = y$, **3 is called base and** x **is index. The logarithm of the number** $y(>0)$ **to the base 3 is** x.

i.e. $3^x = y \Leftrightarrow \log_3 y = x$

In general, if $a^x = y, (a > 0, y > 0)$ **then** x **is called the logarithm of** y **to the base** a **and is written as** $\log_a y = x$

$$\boxed{a^x = y \;\Leftrightarrow\; \log_a y = x, \;\; a, y > 0, \;\; a \neq 1}$$

For example:

$$2^5 = 32 \quad \Leftrightarrow \log_2 32 = 5$$
$$10^3 = 1000 \Leftrightarrow \log_{10} 1000 = 3$$
$$3^{-4} = \frac{1}{81} \quad \Leftrightarrow \log_3 \frac{1}{81} = -4$$
$$\left(\frac{1}{2}\right)^5 = \frac{1}{32} \Leftrightarrow \log_{\frac{1}{2}} \frac{1}{32} = 5$$
$$a^1 = a \quad \Leftrightarrow \log_a a = 1$$

Some fundamental properties of logarithms.

m, n **and** a **are positive numbers and** $a \neq 1$.
(a) $\log_a mn = \log_a m + \log_a n$
(b) $\log_a \frac{m}{n} = \log_a m - \log_a n$
(c) $\log_a m^p = p \log_a m$ **where** p **is rational.**

Let $\log_a m = x$ **and** $\log_a n = y$

$$\log_a m = x \Leftrightarrow m = a^x$$
$$\log_a n = y \Leftrightarrow n = a^y$$

(a) $mn = a^x \times a^y = a^{x+y} \Leftrightarrow \log_a mn = x + y$

$\log_a mn = x + y = \log_a m + \log_a n$

(b) $\frac{m}{n} = \frac{a^x}{a^y} = a^{x-y} \Leftrightarrow \log_a \left(\frac{m}{n}\right) = x - y$

$\log_a \frac{m}{n} = x - y = \log_a m - \log_a n$

(c) $m^p = \left(a^x\right)^p = a^{px}$

$\log_a m^p = px = p \log_a m$

Example 1
Find the values of the following

(a) $\log_{10} 5 - \log_{10} 16 + 2\log_{10} 2 + \log_{10} 8$

(b) $\log_{10} 54 - \log_{10} 15 + 2\log_{10} \frac{5}{3}$

(a) $\log_{10} 5 - \log_{10} 16 + 2\log_{10} 2 + \log_{10} 8$

$= \log_{10} 5 - \log_{10} 16 + \log_{10} 2^2 + \log_{10} 8$

$= \log_{10} \left(\frac{5 \times 2^2 \times 8}{16}\right)$

$= \log_{10} 10 = 1$

(b) $\log_{10} 54 - \log_{10} 15 + 2\log_{10} \frac{5}{3}$

$= \log_{10} 54 - \log_{10} 15 + \log_{10} \frac{5}{3}$

$= \log_{10} 54 - \log_{10} 15 + \log_{10} \left(\frac{5}{3}\right)^2$

$= \log_{10} \left(\frac{54 \times \left(\frac{5}{3}\right)^2}{15} \right)$

$= \log_{10} \left(\frac{54 \times 25}{9 \times 15} \right)$

$= \log_{10} 10 = 1$

Example 2

Solve: (a) $3\log x + \log 96 = 2\log 9 + \log 4$

(b) $4\log x + 6\log 3 = \log 625 + \log 9$

(a) $3\log x + \log 96 = 2\log 9 + \log 4$

$\log x^3 + \log 96 = \log 9^2 + \log 4$

$\log\left(x^3 \times 96\right) = \log\left(9^2 \times 4\right)$

$x^3 \times 96 = 9^2 \times 4$

$x^3 = \frac{9^2 \times 4}{96}$

$x^3 = \frac{27}{8} = \left(\frac{3}{2}\right)^3$

$x = \frac{3}{2}$

(b) $4\log x + 6\log 3 = \log 625 + \log 9$

$\log x^4 + \log 3^6 = \log 625 + \log 9$

$\log\left(x^4 \times 3^6\right) = \log\left(625 \times 9\right)$

$x^4 \times 3^6 = 625 \times 9$

$x^4 = \frac{625 \times 9}{3^6} = \left(\frac{5}{3}\right)^4$

$x = \frac{5}{3}$

Example 3

Given that $\log_{10} 2 = 0.3010$ **and** $\log_{10} 3 = 0.4771$, **find the values of (a)** $\log_{10} 18$ **(b)** $\log_{10} 15$ **(c)** $\log_{10} 0.012$

(a)
$$\begin{aligned}\log_{10} 18 &= \log_{10}(2\times 3^2)\\ &= \log_{10} 2 + \log_{10} 3^2\\ &= \log_{10} 2 + 2\log_{10} 3\\ &= 0.3010 + 2\times 0.4771\\ &= 0.3010 + 0.9542 = 1.2552\end{aligned}$$

(b)
$$\begin{aligned}\log_{10} 15 &= \log_{10}(5\times 3)\\ &= \log_{10} 5 + \log_{10} 3\\ &= \log_{10}\frac{10}{2} + \log_{10} 3\\ &= \log_{10} 10 - \log_{10} 2 + \log_{10} 3\\ &= 1 - 0.3010 + 0.4771\\ &= 1.1761\end{aligned}$$

(c)
$$\begin{aligned}\log_{10} 0.012 &= \log_{10}\frac{12}{1000}\\ &= \log_{10} 12 - \log_{10} 1000\\ &= \log_{10}\left(2^2\times 3\right) - \log_{10} 10^3\\ &= \log_{10} 2^2 + \log_{10} 3 - 3\log_{10} 10\\ &= 2\log_{10} 2 + \log_{10} 3 - 3\log_{10} 10\\ &= 2\times 0.3010 + 0.4771 - 3\\ &= 1.0791 - 3\\ &= 1 + 0.0791 - 3\\ &= -2 + 0.0791 \quad = \bar{2}.0791\end{aligned}$$

Example 4

Find the value of

(a) $\log_{\sqrt{3}} \frac{1}{243}$ **(b)** $\log_{2\sqrt{2}} 16$

Let $\log_{\sqrt{3}} \frac{1}{243} = x$

$\left(\sqrt{3}\right)^{x} = 243$

$3^{\frac{1}{2}x} = 3^{5}$

$\frac{1}{2}x = 5$

$x = 10$

Let $\log_{2\sqrt{2}} 16 = y$

$\left(2\sqrt{2}\right)^{y} = 16$

$\left(2 \times 2^{\frac{1}{2}}\right)^{y} = 2^{4}$

$2^{\frac{3}{2}y} = 2^{4}$

$\frac{3}{2}y = 4$

$y = \frac{8}{3}$

Exercise 5

1 If $x = 27$ **and** $y = 4$ **find the values of**

(a) $\left(x^{\frac{2}{3}}y\right)^{\frac{1}{2}}$ **(b)** $(2xy)^{-\frac{1}{3}}$ **(c)** $\left(\frac{12y}{x}\right)^{\frac{1}{2}}$ **(d)** $\left(x^{\frac{2}{3}} + y^{2}\right)^{-\frac{1}{2}}$

2 Find the values of the following

(a) $\left(25^{\frac{1}{2}} \times 16^{\frac{1}{4}}\right)^{-2}$ **(b)** $\left(\frac{64^{\frac{1}{6}} + 27^{\frac{2}{3}}}{110}\right)^{2}$ **(c)** $\left(\frac{81}{24}\right)^{\frac{1}{3}}$

3 If $x = 81$, $y = 16$ **and** $z = 25$, **find the values of**

(a) $(xy)^{\frac{3}{4}}$ **(b)** $x^{\frac{1}{4}} + y^{\frac{1}{2}}$ **(c)** $\left[\frac{x^{\frac{1}{4}} + y^{\frac{1}{4}}}{z^{-\frac{1}{2}}}\right]^{\frac{1}{2}}$

4 Simplify.

$\left(\frac{4}{9}\right)^{\frac{3}{2}} \times \frac{1}{3^{-4}} \times \left(\frac{27}{8}\right)^{\frac{1}{3}}$

5 Solve:

(a) $3^{x+1} = 243$ **(b)** $16^{x-1} = \frac{1}{8}$

(c) $4^{3x-1} = \left(\frac{1}{2}\right)^{x-1}$ **(d)** $27^{x-3} = 3 \times 9^{x-2}$

(e) $3^{x^{2}} = 9^{x+4}$ **(f)** $9^{x} - 4 \times 3^{x} + 3 = 0$

6 Evaluate each of the following

(i) $\log_{3} 81$ **(ii)** $\log_{3\sqrt{2}} 324$ **(iii)** $\log_{2\sqrt{3}} 144$ **(iv)** $\log_{343} 7$

7 **Find the values of**

(a) $\log_{10}\frac{12}{5}+\log_{10}\frac{25}{21}-\log_{10}\frac{2}{7}$

(b) $\log_{10}\frac{3}{4}+\log_{10}\frac{10}{9}+\log_{10}12-2$

(c) $\frac{\log_{10}8}{\log_{10}4}$

(d) $3\log_{10}2+2\log_{10}5-\log_{10}2$

8 **Find the value of** x **in the following equations**

(a) $5\log x-\log 729=6\log 2+11\log x$

(b) $4\log x+2\log 9=3\log 24-\log 54$

(c) $2\log x=\log 3+\log(2x-3)$

9 **Solve the equations**

(a) $2^{2+2x}+3.2^{x}-1=0$

(b) $\log_{10}(x^2+1)-2\log_{10}x=1$

10 **Show that**

$$\log_{10}2+16\log_{10}\frac{16}{15}+12\log_{10}\frac{25}{24}+7\log_{10}\frac{81}{80}=1$$

11 **Prove that**

(a) $\log(ab^2)-\log(ac)+\log(bc^4)-3\log(bc)=0$ **the base being the same throughout.**

(b) $\log(\log x^5)-\log(\log x^2)=\log\frac{5}{2}$

(c) $\log\left(\frac{a^2}{bc}\right)+\log\left(\frac{b^2}{ca}\right)+\log\left(\frac{c^2}{ab}\right)=0$

12. **If** $a^2+b^2=7ab$, **show that** $\log\left(\frac{a+b}{3}\right)=\frac{1}{2}\log a+\frac{1}{2}\log b$

13. **If** $\log\left(\frac{x+y}{2}\right)=\frac{1}{2}\log x+\frac{1}{2}\log y$, **prove that** $x=y$

14. **Prove that** $\log(1+2+3)=\log 1+\log 2+\log 3$

15. **If** x, y, z **are any consecutive three positive integers, prove that** $\log(1+xz)=2\log y$

16. **Prove that** $\log a+\log a^2+.....+\log a^{2n}=n(2n+1)\log a$, **where** $a>0$.

17. **If** $\log(x+y)=\log x-\log y$, **show that** $x(1-y)=y^2$

18. If $2^x.3^y = 3^x.4^x = 6$, , show that $x^2 - 2y^2 = 2x - 3y$

19. If $\frac{\log x}{2} = \frac{\log y}{3} = \frac{\log z}{5}$, prove that

(i) $xy = z$ and (ii) $x^8 = y^2 z^2$

20. If $\frac{\log x}{1} = \frac{\log y}{3} = \frac{\log z}{5}$, prove that $x^5.y^3.z^{-2} = 1$